

Name: _____

Use the rectangular prism at the right. Think of each edge as part of a line.

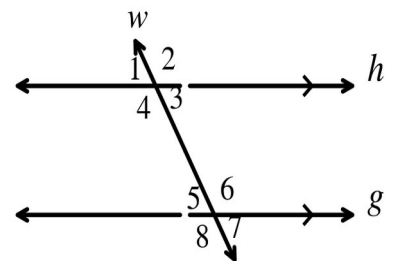
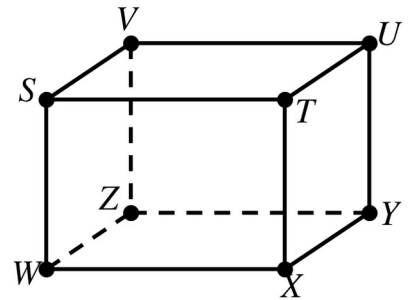
1. Name a segment parallel to ST .
2. Name a segment skew to ST .
3. Name a plane parallel to plane $STXW$.

Use the second diagram, in which $g \parallel h$.

4. Name a pair of alternate exterior angles.
5. Name a pair of consecutive interior angles.
6. $m\angle 1 = 109^\circ$. Find $m\angle 5$.
7. $m\angle 4 = 71^\circ$. Find $m\angle 6$.
8. $m\angle 3 = 124^\circ$. Find $m\angle 6$.

Two parallel lines are cut by a transversal. Find x .

9. A pair of corresponding angles measure $(4x + 7)^\circ$ and 63° .
10. A pair of consecutive interior angles measure $(5x - 5)^\circ$ and 100° .



Answer each question and name the theorem that justifies it.

- 11.** In the second diagram, $\angle 2 \cong \angle 6$. What can you conclude?
- 12.** In the second diagram, $\angle 3$ and $\angle 6$ are supplementary. What can you conclude?
- 13.** In a plane, line $a \perp$ line c and line $b \perp$ line c .
- 14.** In a plane, line $r \parallel$ line s and line $t \perp$ line r .
- 15.** Two lines intersect to form a linear pair of congruent angles.

Slopes and equations of lines.

- 16.** Find the slope of the line through $(-5, 4)$ and $(3, -2)$.
- 17.** Write an equation of the line through $(1, -3)$ that is parallel to $y = 5x + 2$.
- 18.** Write an equation of the line through $(-2, 7)$ that is perpendicular to $y = (1/3)x - 4$.
- 19.** Are the lines parallel, intersecting, or coinciding? $y = -4x + 6$ and $8x + 2y = 12$
- 20.** Segment $\overline{AB}$ has endpoints $A(-1, 5)$ and $B(7, 5)$. Write an equation of its perpendicular bisector.

